Skill - 1

Encapsulation Builder – Wireshark Packet Analysis

2520030123

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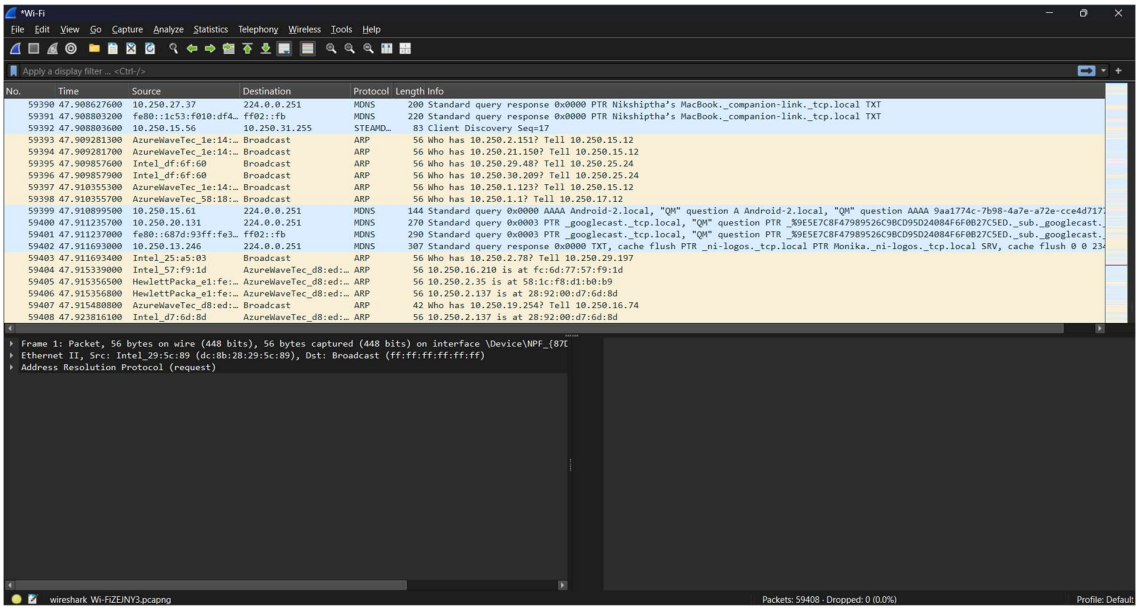
Sec- 2

Aim

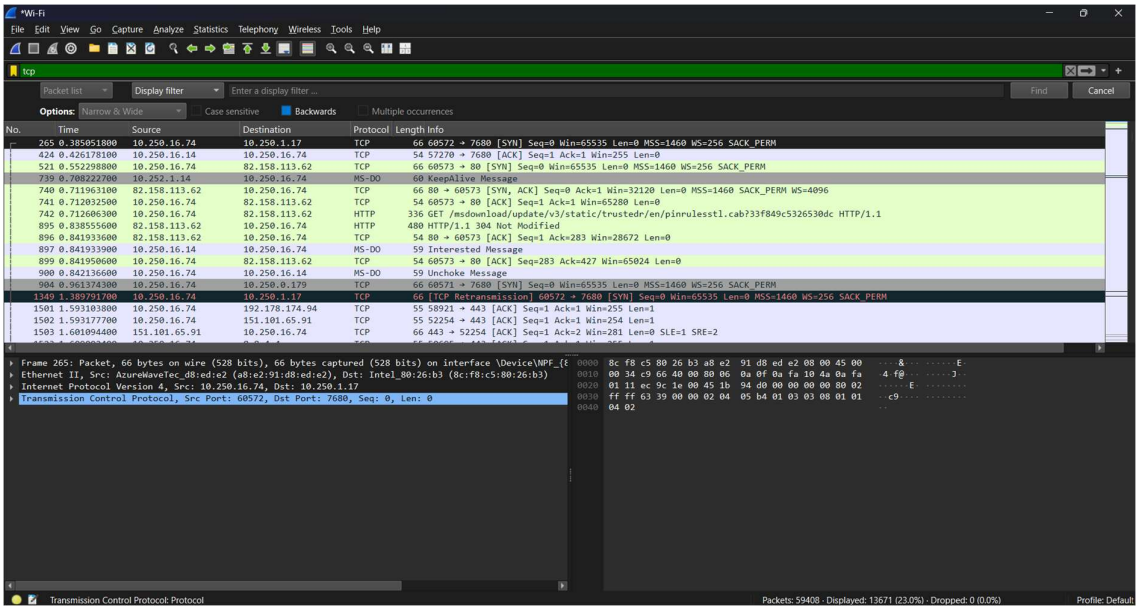
To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

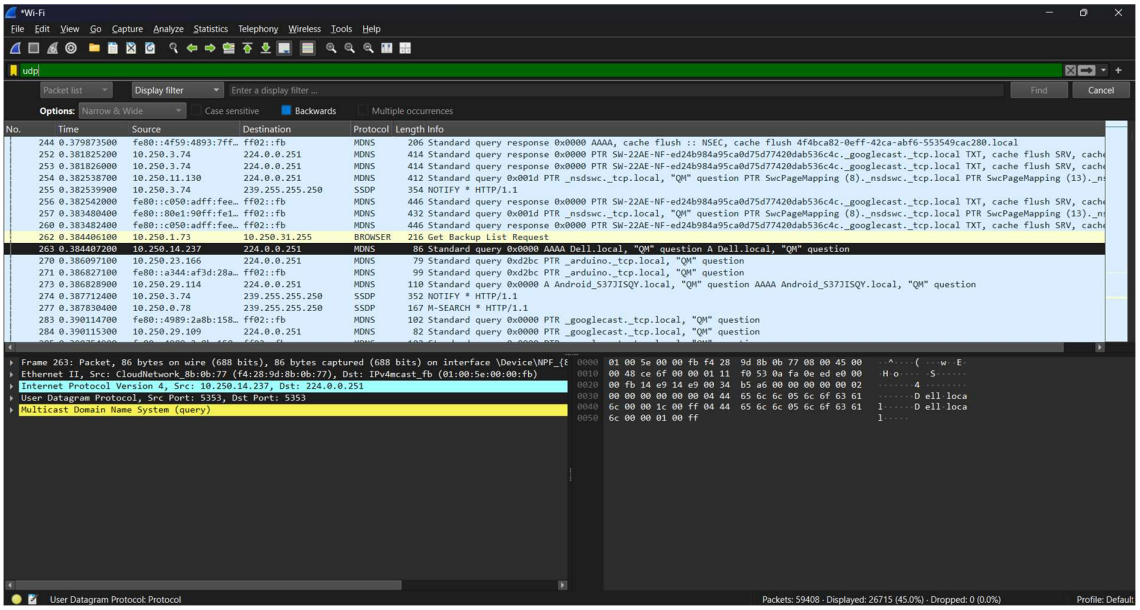
Step 1 & 2 : Open Wireshark and start capturing on Wi-Fi interface

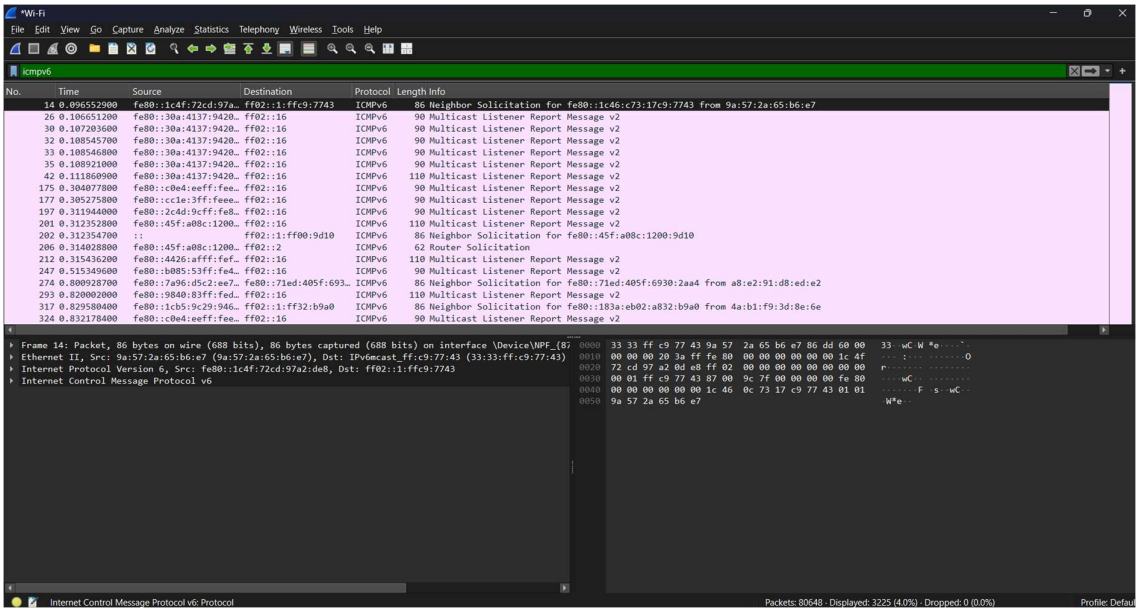
Step 3 : Browse a few websites then stop the capture.



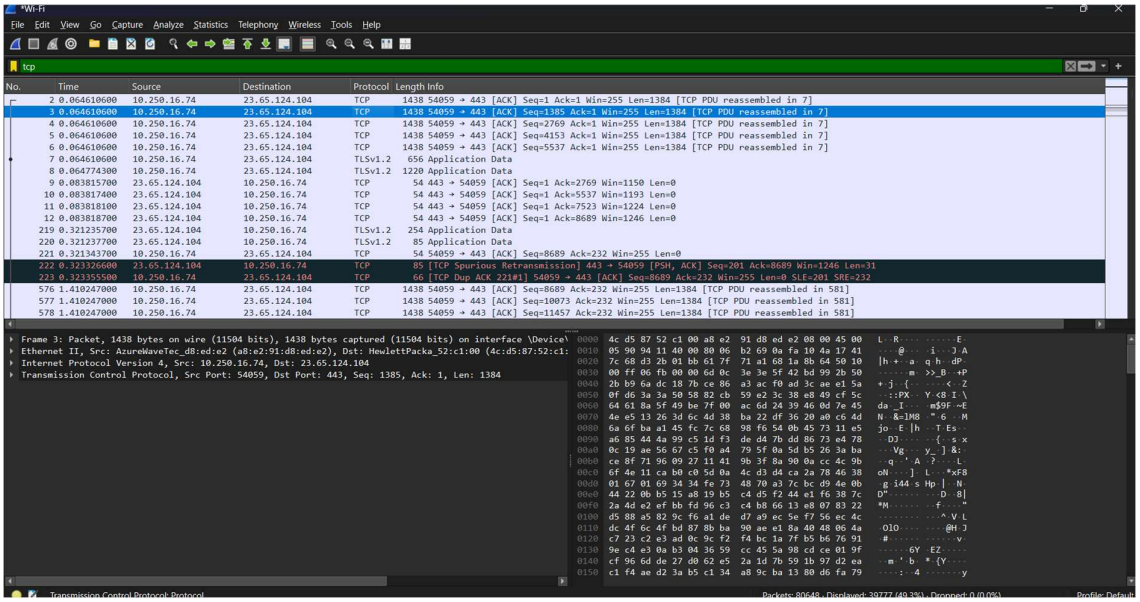
Step 4 : Apply the display filters tcp , udp, icmp/icmpv6


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
Packet list
Display filter
Enter a display filter ...
Find
Cancel
Options: Narrow & Wide
Case sensitive
Backwards
Multiple occurrences
No. Time Source Destination Protocol Length Info
265 0.385051800 10.250.16.74 10.250.1.17 TCP 66 60572 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
424 0.426178100 10.250.16.14 10.250.16.74 TCP 54 57270 → 7680 [ACK] Seq=1 Ack=1 Win=255 Len=0
521 0.552298800 10.250.16.74 82.158.113.62 TCP 66 60573 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
739 0.708222700 10.252.1.14 10.250.16.74 MS-DO 60 KeepAlive Message
740 0.711963100 82.158.113.62 10.250.16.74 TCP 66 80 → 60573 [SYN, ACK] Seq=0 Ack=1 Win=32120 Len=0 MSS=1460 SACK_PERM WS=4096
741 0.712032500 10.250.16.74 82.158.113.62 TCP 54 60573 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
742 0.712606300 10.250.16.74 82.158.113.62 HTTP 336 GET /msdownload/update/v3/static/trustedr/en/pinrulesstl.cab?33f849c5326530dc HTTP/1.1
895 0.838555600 82.158.113.62 10.250.16.74 HTTP 480 HTTP/1.1 304 Not Modified
896 0.841933600 82.158.113.62 10.250.16.74 TCP 54 80 → 60573 [ACK] Seq=1 Ack=283 Win=28672 Len=0
897 0.841933900 10.250.16.14 10.250.16.74 MS-DO 59 Interested Message
899 0.841950600 10.250.16.74 82.158.113.62 TCP 54 60573 → 80 [ACK] Seq=283 Ack=427 Win=65024 Len=0
900 0.842136600 10.250.16.74 10.250.16.14 MS-DO 59 Unchoke Message
904 0.961374300 10.250.16.74 10.250.0.179 TCP 66 60571 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1349 1.389791700 10.250.16.74 10.250.1.17 TCP 66 [TCP Retransmission] 60572 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1501 1.593103800 10.250.16.74 192.178.174.94 TCP 55 58921 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
1502 1.593177700 10.250.16.74 151.101.65.91 TCP 55 52254 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
1503 1.601094400 151.101.65.91 10.250.16.74 TCP 66 443 → 52254 [ACK] Seq=1 Ack=281 Len=0 SLE=1 SRE=2
Frame 265: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{E
Ethernet II, Src: AzureWaveTec_d8:ed:e2 (a8:e2:91:d8:ed:e2), Dst: Intel_80:26:b3 (8c:f8:c5:80:26:b3)
Internet Protocol Version 4, Src: 10.250.16.74, Dst: 10.250.1.17
Transmission Control Protocol, Src Port: 60572, Dst Port: 7680, Seq: 0, Len: 0
0000 8c f8 c5 80 26 b3 a8 e2 91 d8 ed e2 08 00 45 00 ····&····· ·····E·
0010 00 34 c9 66 40 00 80 06 0a 0f 0a fa 10 4a 0a fa ·4·f@··· ·····J··
0020 01 11 ec 9c 1e 00 45 1b 94 d0 00 00 00 00 80 02 ·····E· ········
0030 ff ff 63 39 00 00 02 04 05 b4 01 03 03 08 01 01 ··c9···· ········
0040 04 02 ··
Transmission Control Protocol: Protocol
Packets: 59408 · Displayed: 13671 (23.0%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Packet list
Display filter
Enter a display filter ...
Find
Cancel
Options: Narrow & Wide
Case sensitive
Backwards
Multiple occurrences
No. Time Source Destination Protocol Length Info
244 0.379873500 fe80::4f59:4893:7ff… ff02::fb MDNS 206 Standard query response 0x0000 AAAA, cache flush :: NSEC, cache flush 4f4bca82-0eff-42ca-abf6-553549cac280.local
252 0.381825200 10.250.3.74 224.0.0.251 MDNS 414 Standard query response 0x0000 PTR SW-22AE-NF-ed24b984a95ca0d75d77420dab536c4c._googlecast._tcp.local TXT, cache flush SRV, cache
253 0.381826000 10.250.3.74 224.0.0.251 MDNS 414 Standard query response 0x0000 PTR SW-22AE-NF-ed24b984a95ca0d75d77420dab536c4c._googlecast._tcp.local TXT, cache flush SRV, cache
254 0.382538700 10.250.11.130 224.0.0.251 MDNS 412 Standard query 0x001d PTR _nsdswc._tcp.local, "QM" question PTR SwcPageMapping (8)._nsdswc._tcp.local PTR SwcPageMapping (13)._n
255 0.382539900 10.250.3.74 239.255.255.250 SSDP 354 NOTIFY * HTTP/1.1
256 0.382542000 fe80::c050:adff:fee… ff02::fb MDNS 446 Standard query response 0x0000 PTR SW-22AE-NF-ed24b984a95ca0d75d77420dab536c4c._googlecast._tcp.local TXT, cache flush SRV, cache
257 0.383480400 fe80::80e1:90ff:fe1… ff02::fb MDNS 432 Standard query 0x001d PTR _nsdswc._tcp.local, "QM" question PTR SwcPageMapping (8)._nsdswc._tcp.local PTR SwcPageMapping (13)._n
260 0.383482400 fe80::c050:adff:fee… ff02::fb MDNS 446 Standard query response 0x0000 PTR SW-22AE-NF-ed24b984a95ca0d75d77420dab536c4c._googlecast._tcp.local TXT, cache flush SRV, cache
262 0.384406100 10.250.1.73 10.250.31.255 BROWSER 216 Get Backup List Request
263 0.384407200 10.250.14.237 224.0.0.251 MDNS 86 Standard query 0x0000 AAAA Dell.local, "QM" question A Dell.local, "QM" question
270 0.386097100 10.250.23.166 224.0.0.251 MDNS 79 Standard query 0xd2bc PTR _arduino._tcp.local, "QM" question
271 0.386827100 fe80::a344:af3d:28a… ff02::fb MDNS 99 Standard query 0xd2bc PTR _arduino._tcp.local, "QM" question
273 0.386828900 10.250.29.114 224.0.0.251 MDNS 110 Standard query 0x0000 A Android_S37JISQY.local, "QM" question AAAA Android_S37JISQY.local, "QM" question
274 0.387712400 10.250.3.74 239.255.255.250 SSDP 352 NOTIFY * HTTP/1.1
277 0.387830400 10.250.0.78 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
283 0.390114700 fe80::4989:2a8b:158… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
284 0.390115300 10.250.29.109 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
Frame 263: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{E
Ethernet II, Src: CloudNetwork_8b:0b:77 (f4:28:9d:8b:0b:77), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Internet Protocol Version 4, Src: 10.250.14.237, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
0000 01 00 5e 00 00 fb f4 28 9d 8b 0b 77 08 00 45 00 ··^····(···w··E·
0010 00 48 ce 6f 00 00 01 11 f0 53 0a fa 0e ed e0 00 ·H·o···· ·S····
0020 00 fb 14 e9 14 e9 00 34 b5 a6 00 00 00 00 00 02 ·····4 ····
0030 00 00 00 00 00 00 04 44 65 6c 6c 05 6c 6f 63 61 ·····D ell·loca
0040 6c 00 00 1c 00 ff 04 44 65 6c 6c 05 6c 6f 63 61 l····· D ell·loca
0050 6c 00 00 01 00 ff l····
User Datagram Protocol: Protocol
Packets: 59408 · Displayed: 26715 (45.0%) · Dropped: 0 (0.0%)
Profile: Default



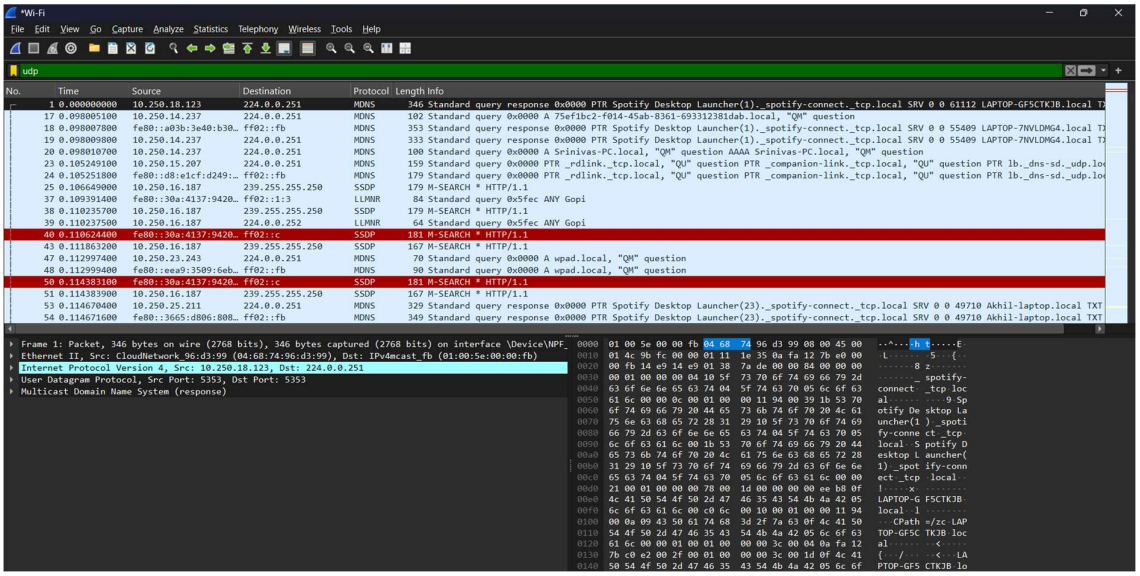
Step 5 : Select a tcp packet and expand the Ethernet II and Internet protocol .

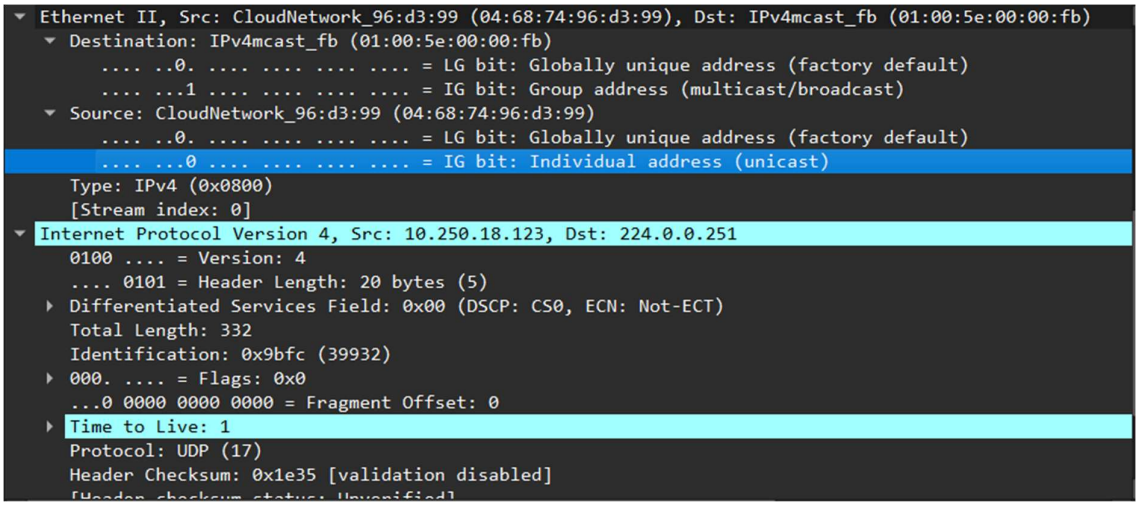


```
Ethernet II, Src: AzureWaveTec_d8:ed:e2 (a8:e2:91:d8:ed:e2), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c
  Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: AzureWaveTec_d8:ed:e2 (a8:e2:91:d8:ed:e2)
  Type: IPv4 (0x0800)
  [Stream index: 1]
Internet Protocol Version 4, Src: 10.250.16.74, Dst: 23.65.124.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 1424
  Identification: 0x9411 (37905)
  010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0xb269 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.250.16.74
```

```
Internet Protocol Version 4, Src: 10.250.16.74, Dst: 23.65.124.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 1424
  Identification: 0x9411 (37905)
  010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0xb269 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.250.16.74
  Destination Address: 23.65.124.104
  [Stream index: 1]
Transmission Control Protocol, Src Port: 54059, Dst Port: 443, Seq: 1385, Ack: 1, Len: 1384
  Source Port: 54059
  Destination Port: 443
  [Stream index: 0]
  [Stream Packet Number: 2]
```

Step 6 : Do the same with udp





Step 7: Observation Table

Parameter	TCP Packet	UDP Packet
Frame Length	1438 bytes	346 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	1424 bytes	332 bytes
Transport Segment Length	1404 bytes	312 bytes
Payload Length	1384 bytes	304 bytes

Answers to the Questions

1. What is encapsulation?

Encapsulation is the process of adding headers to data at each network layer before sending it over a network.

2. What is the purpose of the Ethernet header?

It contains the source and destination MAC addresses and helps deliver data within a local network.

3. What information is stored in the IP header?

It stores the source IP address, destination IP address, packet length, TTL and protocol information.

4. What is the difference between TCP and UDP headers?

TCP has a minimum header size of 20 bytes and provides reliable communication, while UDP has an 8-byte header and is faster with less overhead.

5. Why does each protocol add a header?

Each protocol adds a header to provide the information needed for proper data transmission and delivery.

6. What is packet overhead?

Packet overhead is the extra data added by headers and trailers apart from the actual application data.

7. How does header overhead affect network efficiency?

More header overhead means more bytes are used for control information, leaving less space for actual data and reducing network efficiency.